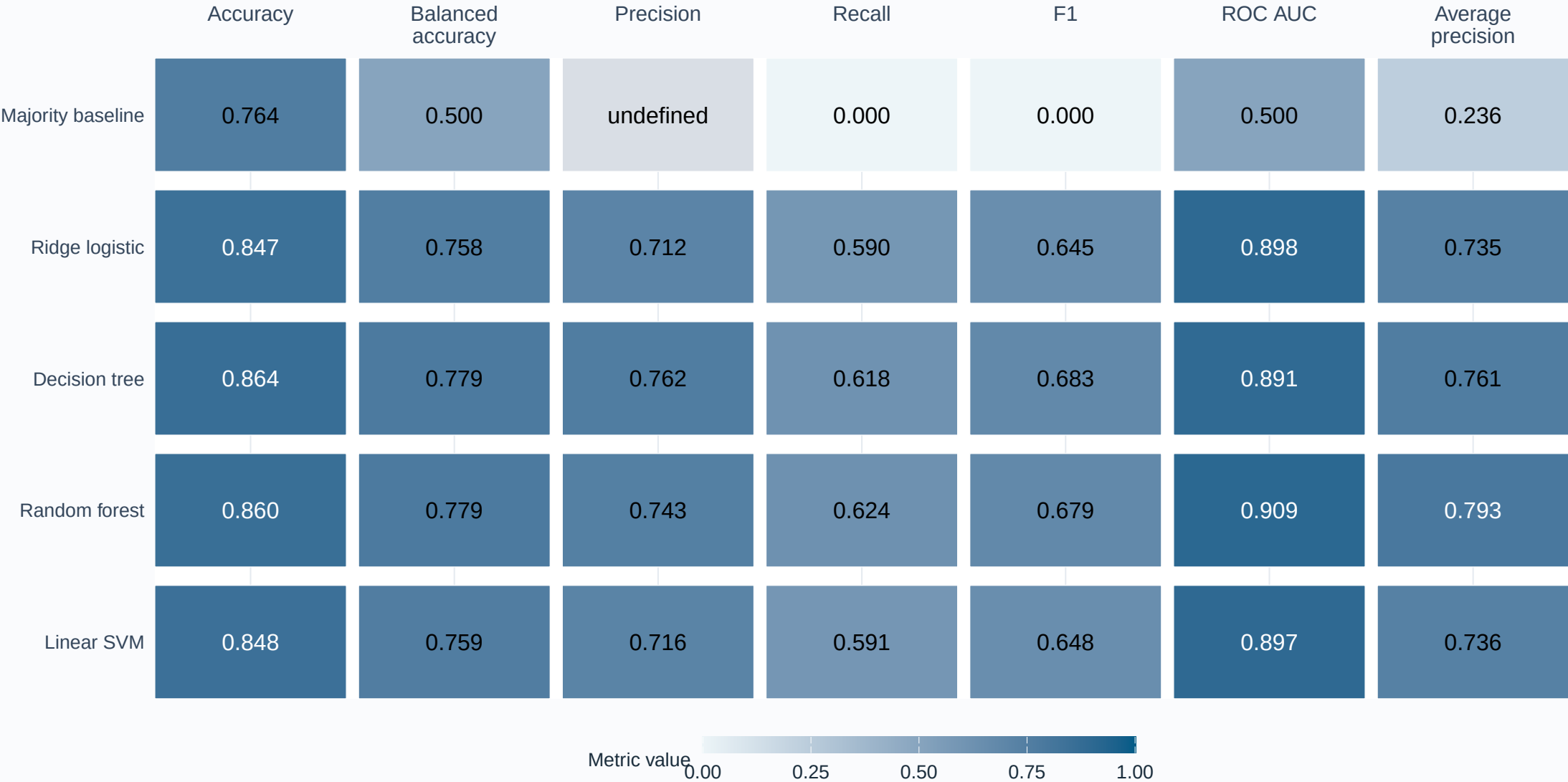


Seven metrics give complementary views of performance

Positive class: >50K | Every cell is a separate metric; higher is better within each column



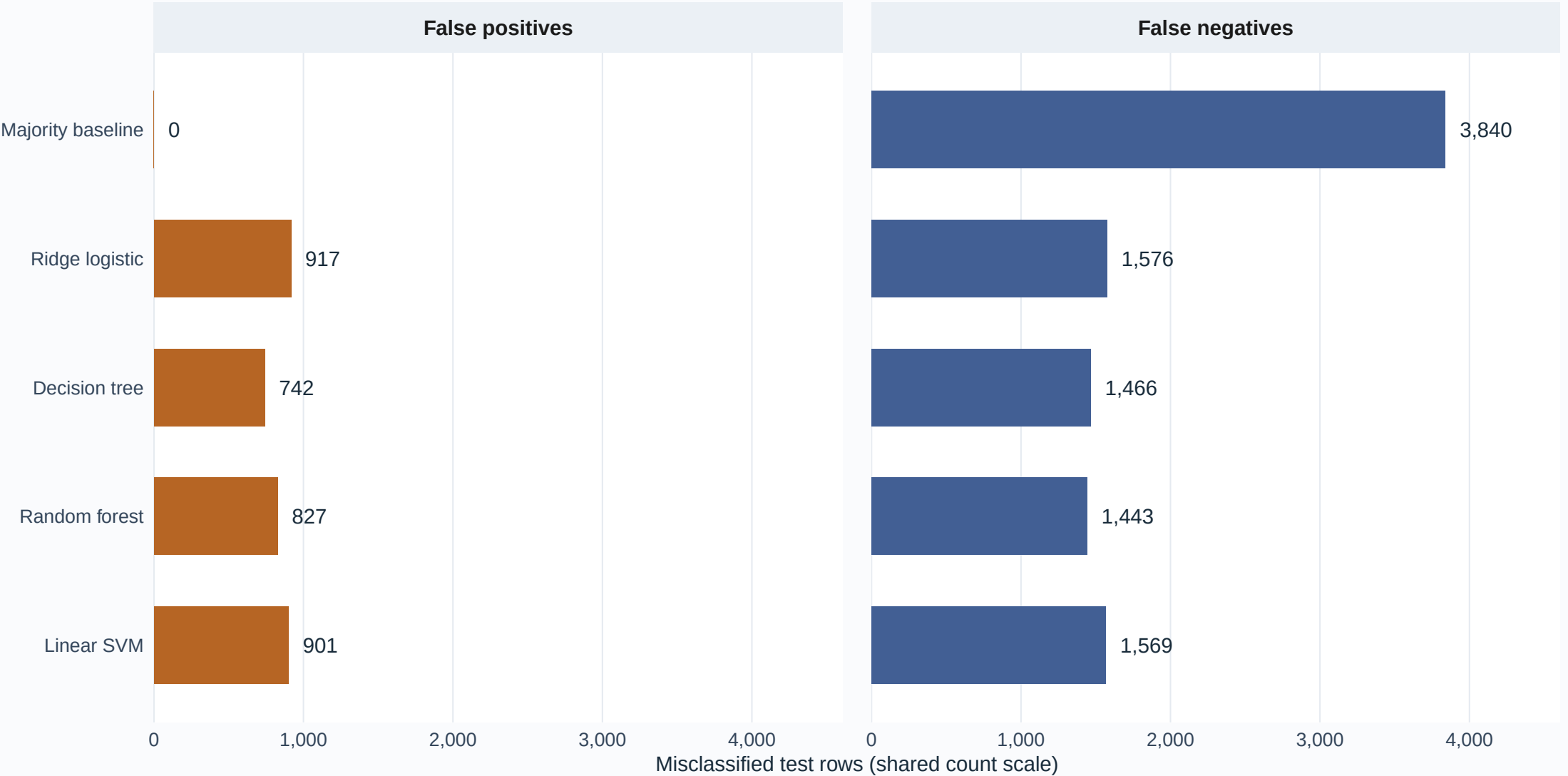
UCI Adult | reused historical benchmark | 16,255 test rows

Point estimates only. The shared 0–1 color scale does not make different metrics interchangeable; no composite score is calculated.

Gray: majority–baseline precision is undefined because it predicts no >50K rows. Threshold and ranking metrics answer different questions.

False negatives outnumber false positives in every model

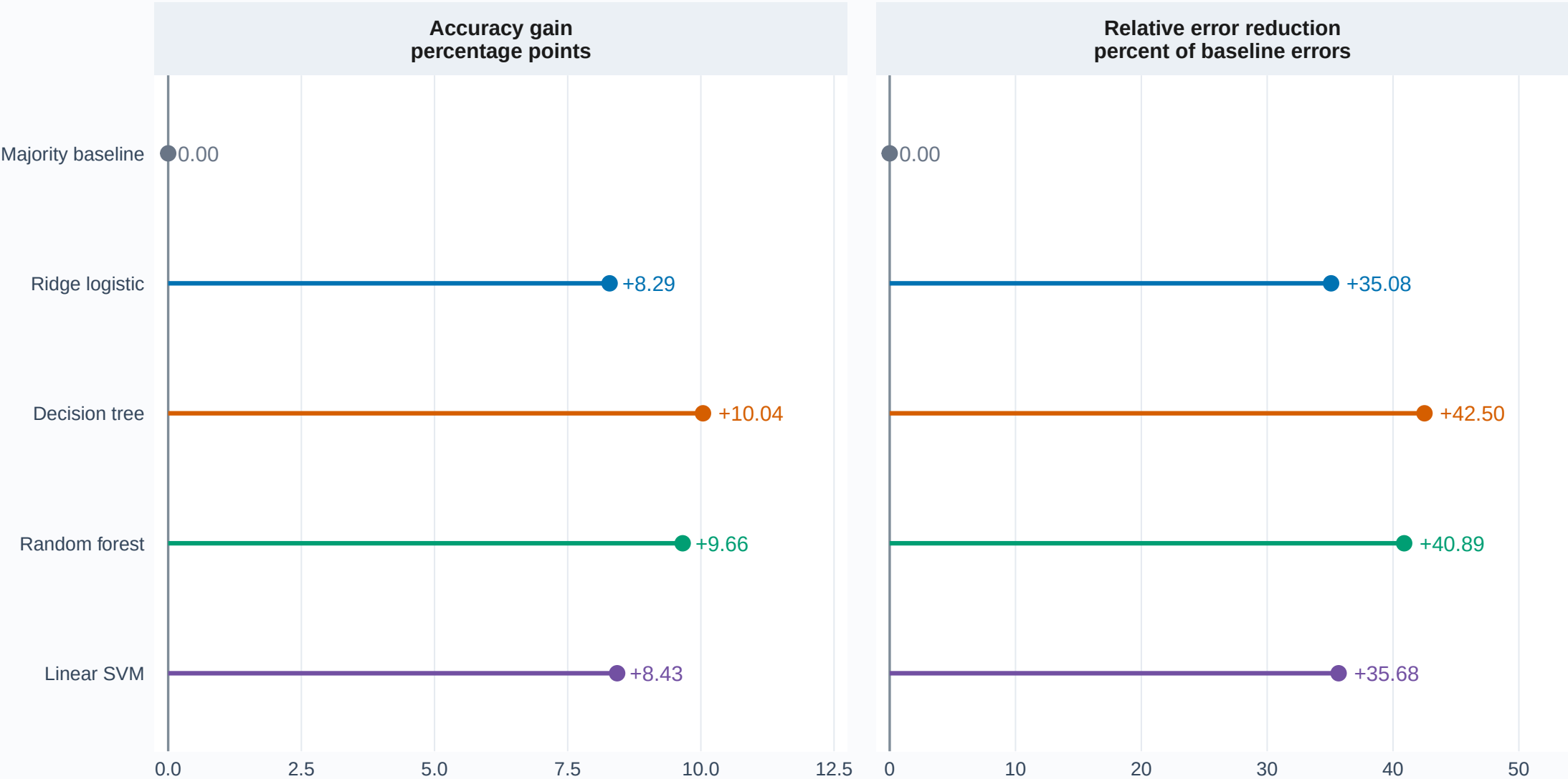
False positive: $\leq 50K$ predicted $> 50K$ | False negative: $> 50K$ predicted $\leq 50K$



UCI Adult | reused historical benchmark | 16,255 test rows
All five models use the same 16,255 test rows: 3,840 actual positives and 12,415 actual negatives. Counts are not error rates.
Decisions use the saved fixed thresholds. The majority baseline has zero false positives and misses every actual positive.

Every trained model improves on the majority baseline

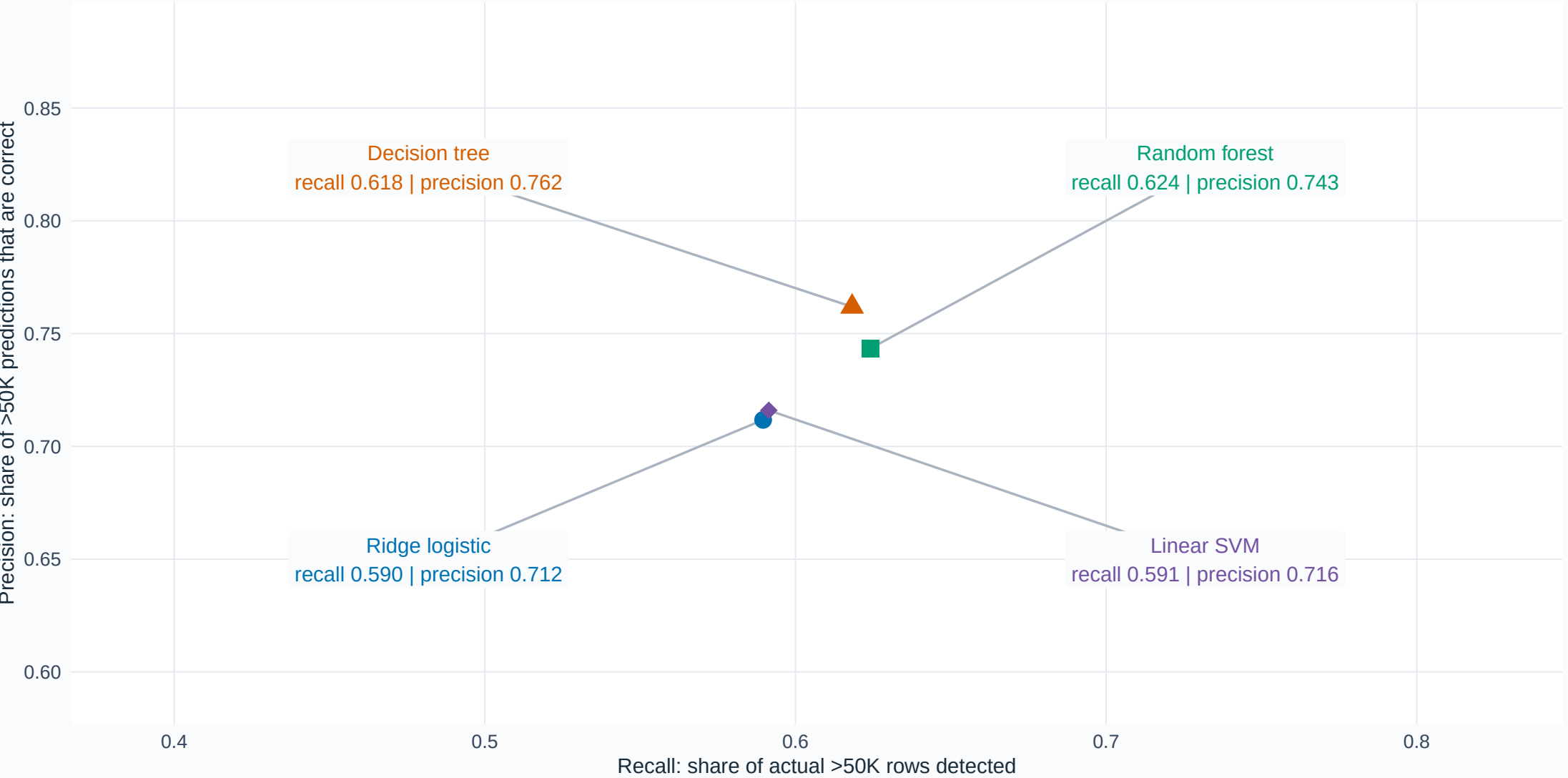
Each panel uses its own stated units; zero means no improvement over the majority baseline



UCI Adult | reused historical benchmark | 16,255 test rows
Accuracy gain = 100 x (model accuracy – baseline accuracy). Relative error reduction = 100 x (baseline errors – model errors) / baseline errors.
Descriptive point differences on the reused test sample. These panels add no uncertainty intervals or significance claims.

Precision and recall describe the saved decisions

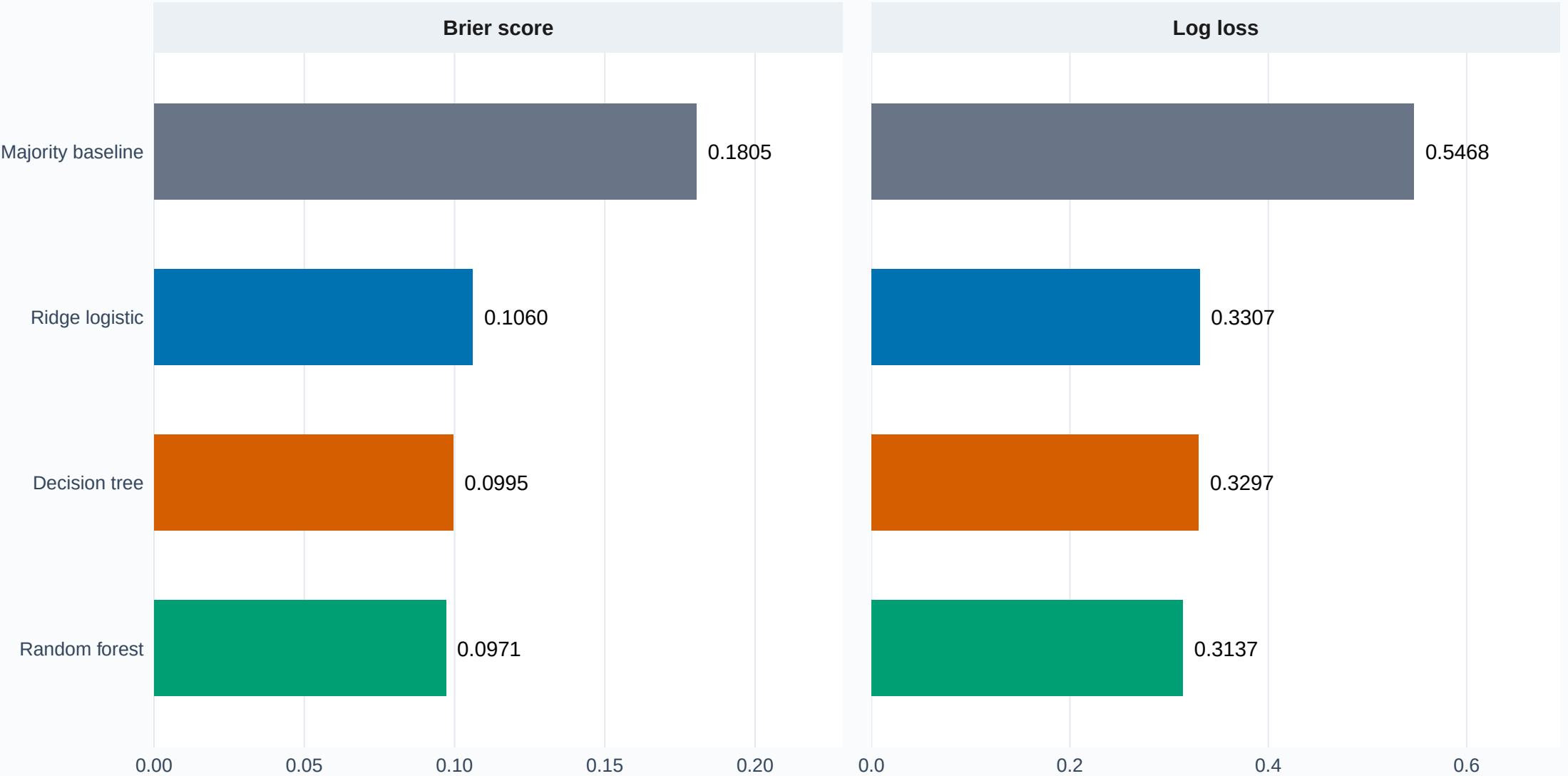
Fixed thresholds: probability 0.5; linear SVM margin 0 | Axes focus on the observed region



UCI Adult | reused historical benchmark | 16,255 test rows
Each point is one fitted model at its saved threshold, not a threshold–search curve. Higher and further right is better on these two metrics.
Majority baseline omitted from the scatter: recall = 0, precision undefined (no positive predictions). No threshold is tuned here.

Probability quality improves beyond the constant baseline

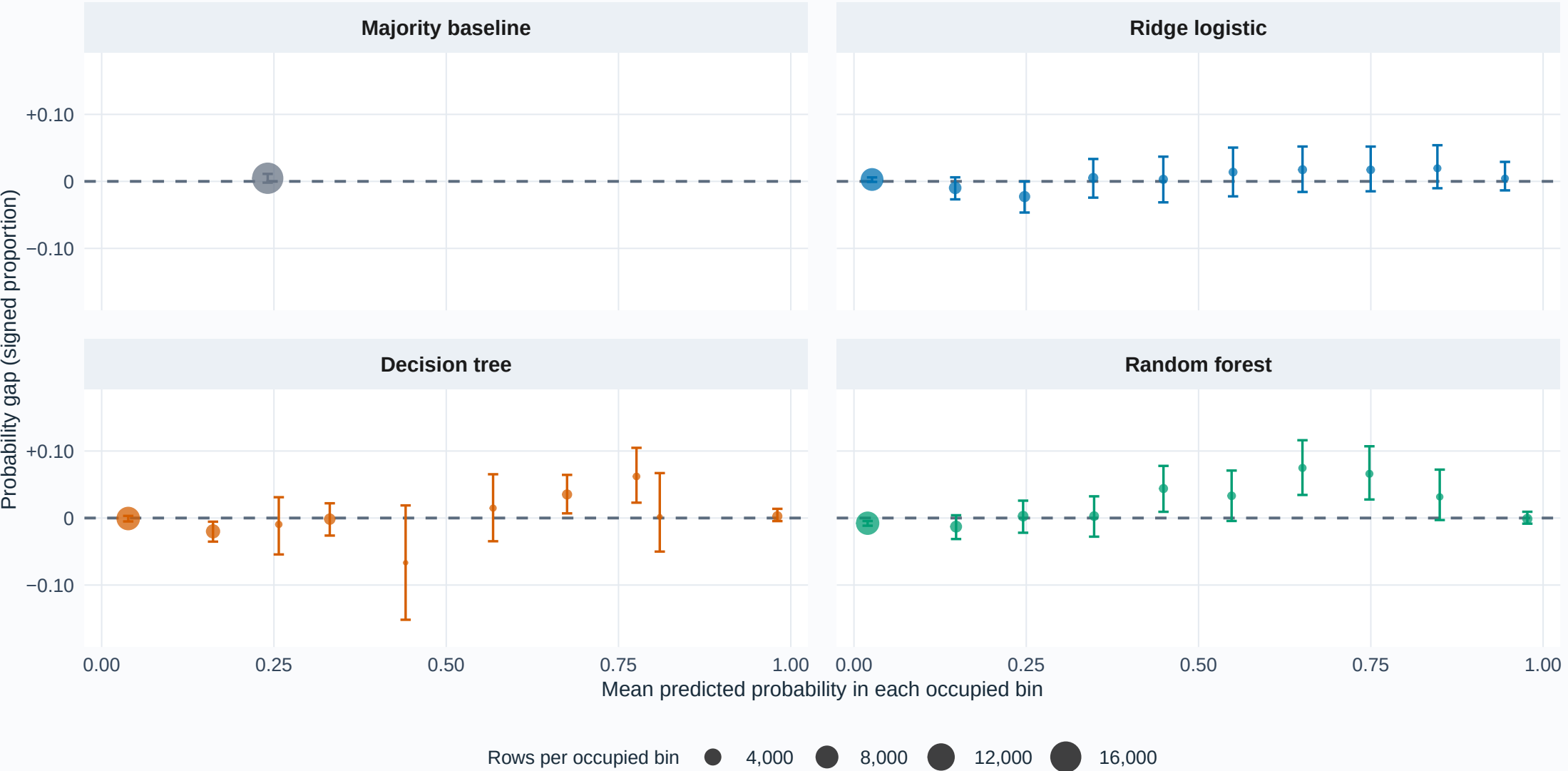
Lower is better in each panel | Both axes start at zero; the two scores have different units



UCI Adult | reused historical benchmark | 16,255 test rows
Brier score and log loss measure overall probability quality, including discrimination and calibration. SVM margins are not probabilities.
Log loss uses clipping to [1e-15, 1 - 1e-15]. Rows clipped: Decision tree 127; Random forest 2,156. Brier scores use original probabilities.

Calibration gaps show overprediction and underprediction

Gap = mean predicted probability – observed >50K rate | Positive = overprediction; negative = underprediction



UCI Adult | reused historical benchmark | 16,255 test rows
Ten fixed-width bins; empty bins omitted. Bars reflect saved 95% Wilson rate intervals; they assume independent rows and hold predictions fixed.
Zero marks agreement. Baseline agreement reflects prevalence, not discrimination. SVM excluded; no recalibration is fitted on this test set.

All 13 trained configurations in the validation search

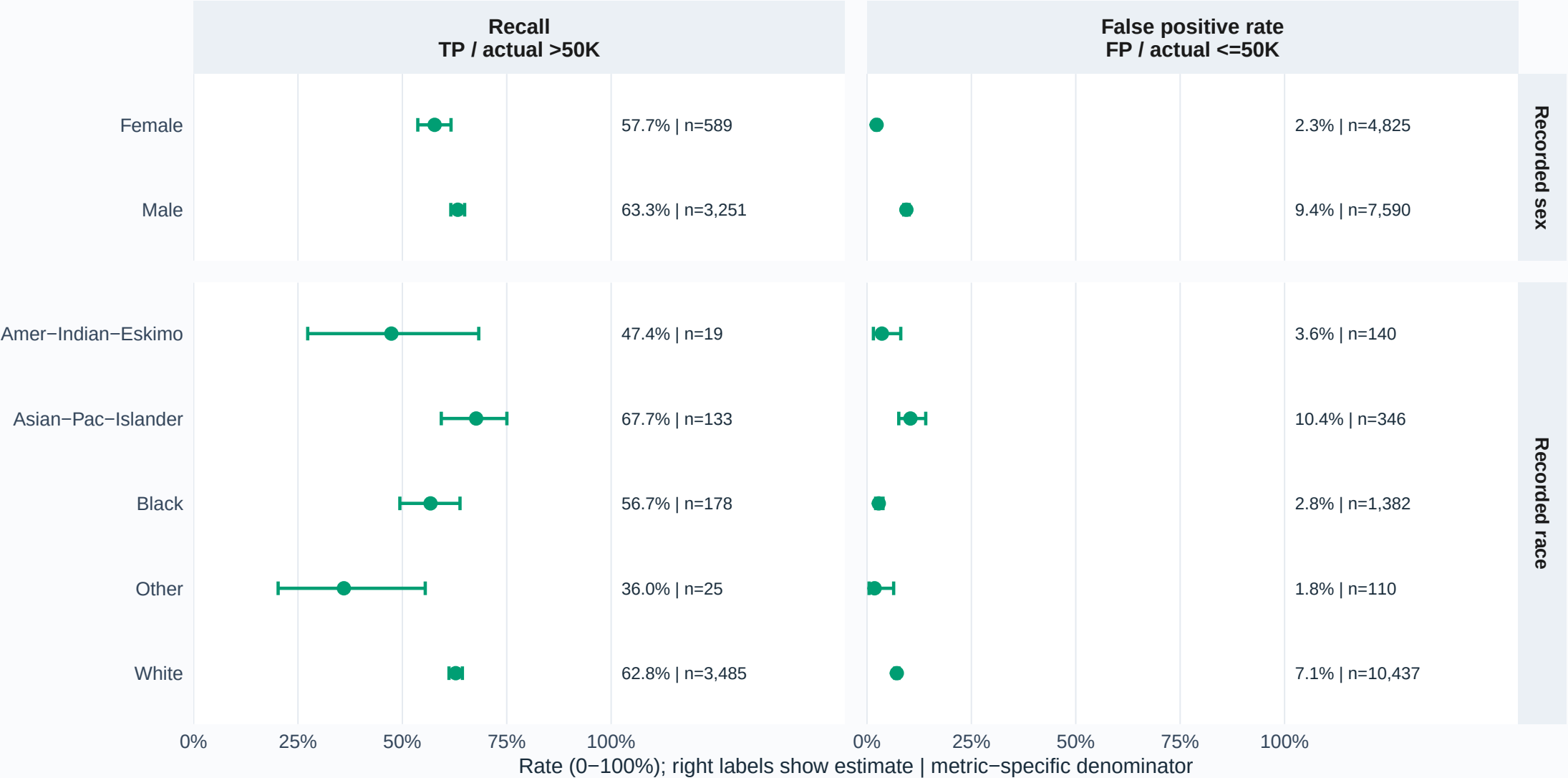
Training folds only | Circles: five fold scores; diamonds and right labels: arithmetic means



UCI Adult | reused historical benchmark | 16,255 test rows
This chart uses training cross-validation, not the 16,255-row test set. Filled diamonds and [selected] identify each family's chosen configuration.
Fold spread is descriptive, not a confidence interval. Selection reuses these folds; tested grid boundaries do not establish a global optimum.

Random forest errors vary across historical groups

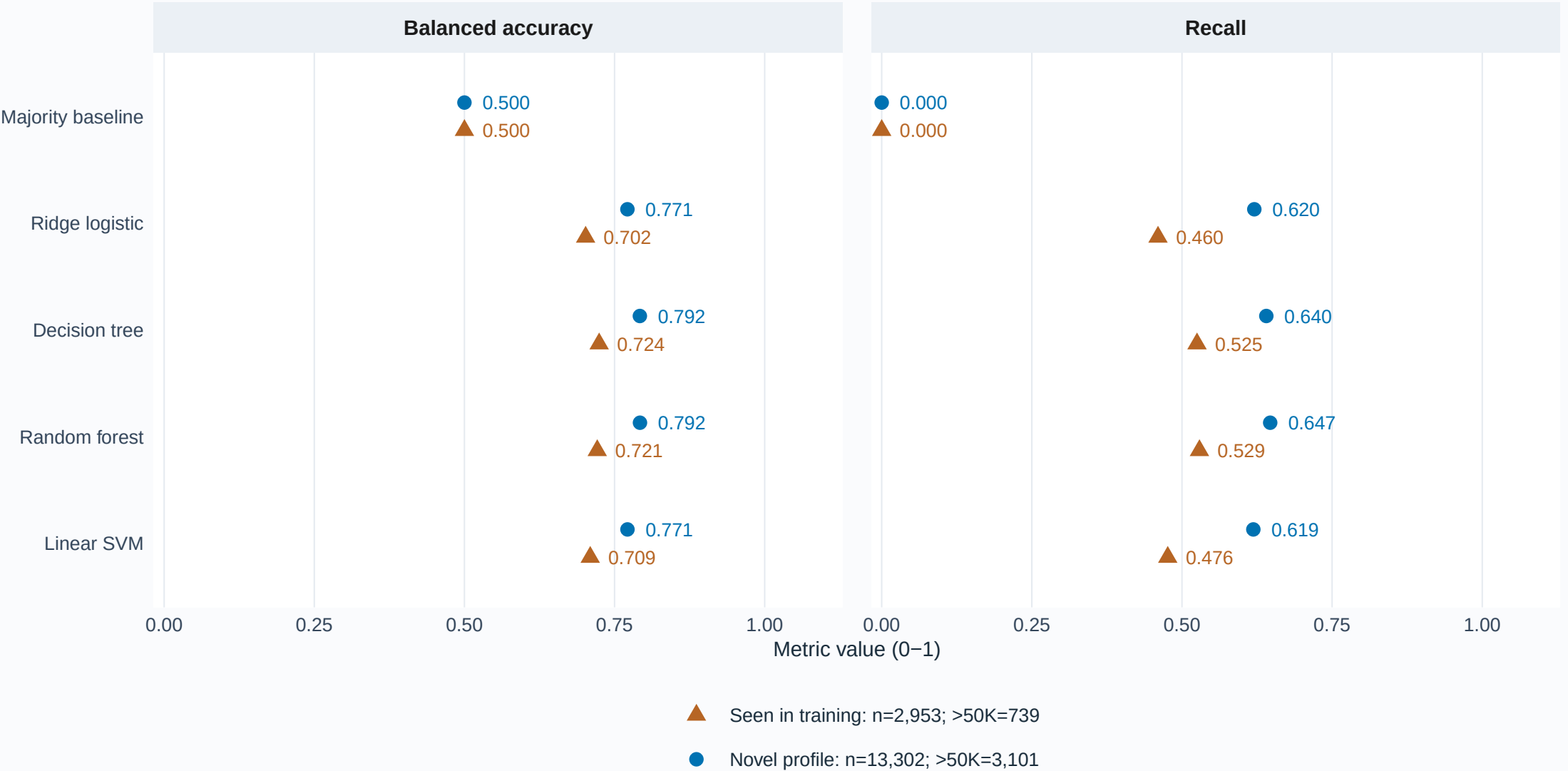
Model family preferred by training CV | Points: test estimates; bars: saved 95% Wilson intervals



UCI Adult | reused historical benchmark | 16,255 test rows
Recall denominators are actual >50K rows; false-positive-rate denominators are actual <=50K rows. Wide bars expose uncertainty in small groups.
Original historical categories retained. Descriptive, not fairness certification; independent-row Wilson intervals are not multiplicity-adjusted.

Test results differ between seen and novel model profiles

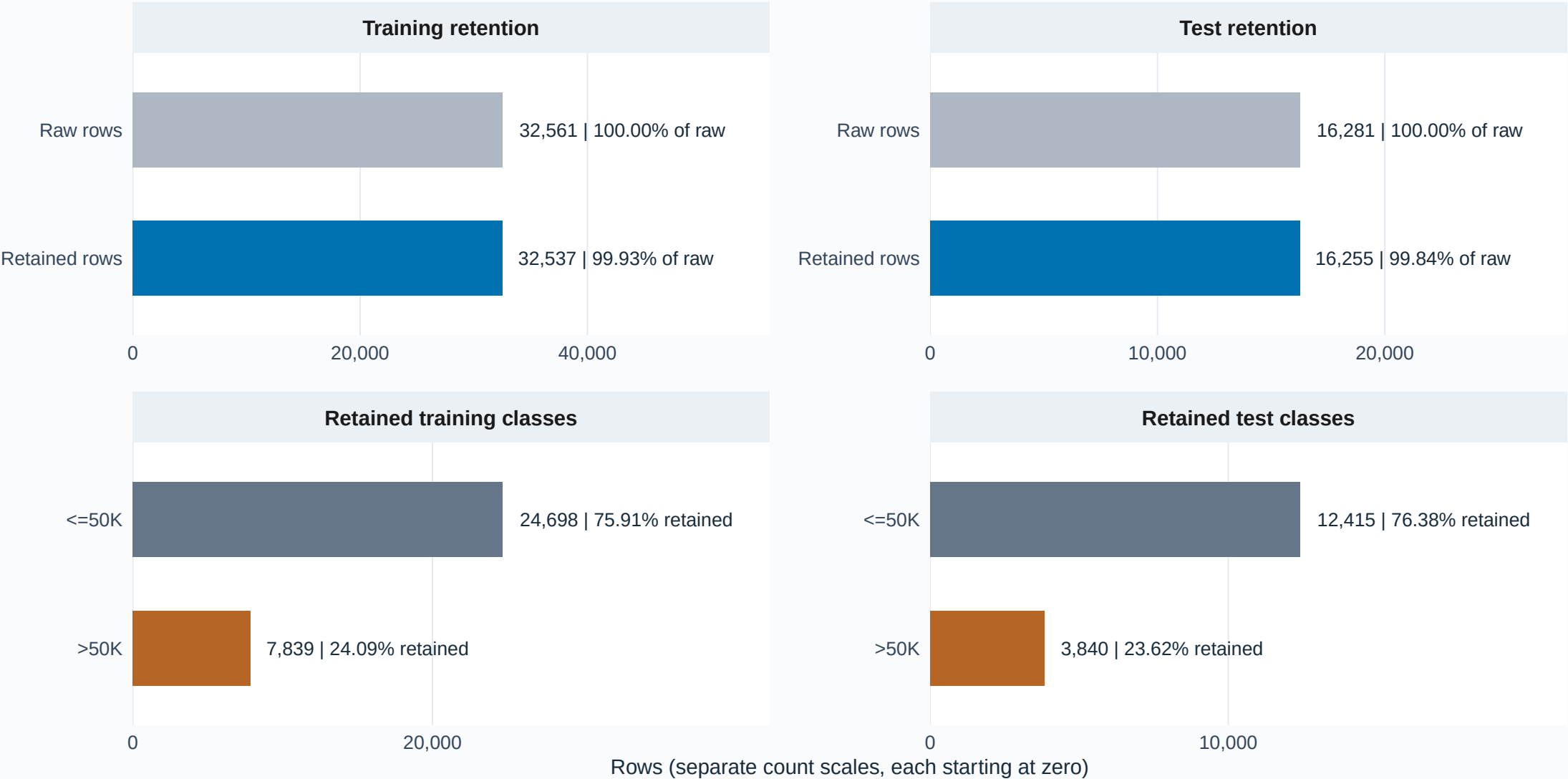
All five models | Same fitted models and fixed thresholds; two descriptive test subsets



UCI Adult | reused historical benchmark | 16,255 test rows
Seen means an identical model-input predictor profile occurs in training. It does not establish that rows identify the same person.
The subsets have different prevalence and case mixes. These point estimates do not isolate an overlap effect; no new intervals are inferred.

Sample retention and class balance define the denominators

Exclusions: 24 repeated labeled training rows; 26 test rows sharing raw predictor signatures with training



UCI Adult | reused historical benchmark | 16,255 test rows
Retention percentages use each raw partition as denominator. Income percentages use each retained partition; the panels are not additive.
Missing values are handled by fold-fitted preprocessing, not complete-case deletion. Profile matches are not confirmed person identities.